2.1 Transformations of Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.1 · Student sheet

I can…

- describe transformations of $f(x) = x^2$.
- find the vertex of a transformed quadratic function and graph it.
- write rules for quadratic functions after transformations.

Vocabulary · fill in the missing word

Quadratic function — $f(x) =$ _____, where $a \neq 0$

Parabola — the U-shaped graph of a _____ function

Vertex form — $f(x) =$ _____

Vertex — (h, k), the lowest or _____ point

Part 1 · Slider lab

Graph $y = a(x - h)^2 + k$ in Desmos with sliders. Describe what each one does.

Slider	What happens as it changes?
a (try $a > 1$, $0 < a < 1$, $a < 0$)	
h	
k	

Part 2 · Notes: fill in the blanks

$f(x) = a(x - h)^2 + k$ has vertex _____.

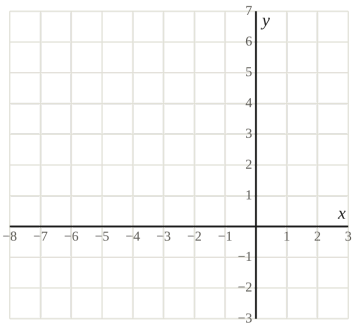
$a < 0$: the parabola _____. $|a| > 1$: _____.

$h > 0$ moves the graph _____; $k > 0$ moves it _____.

Part 3 · Examples (work them with the class)

Example 1

Describe the transformation of $f(x) = x^2$ represented by $g(x) = (x + 4)^2 - 2$. Then graph g.



Example 2

Describe the transformation of $f(x) = x^2$ represented by $g(x) = -\frac{1}{2}x^2 + 3$. Give the vertex.

Example 3

Let g be a vertical stretch by a factor of 3 of $f(x) = x^2$, followed by a translation 1 unit right and 5 units up. Write a rule for g.

Example 4

The height h (in feet) of a kicked ball is $h(x) = -0.03(x - 14)^2 + 6$, where x is the horizontal distance (in feet). Describe the transformation of $f(x) = x^2$. What is the ball's maximum height?

Part 4 · Partner practice

Partner 1

Describe the transformation of $f(x) = x^2$ represented by $g(x) = 2(x - 3)^2$. Give the vertex.

Partner 2

Write a rule for g, a reflection in the x-axis of $f(x) = x^2$ followed by a translation 4 units down.